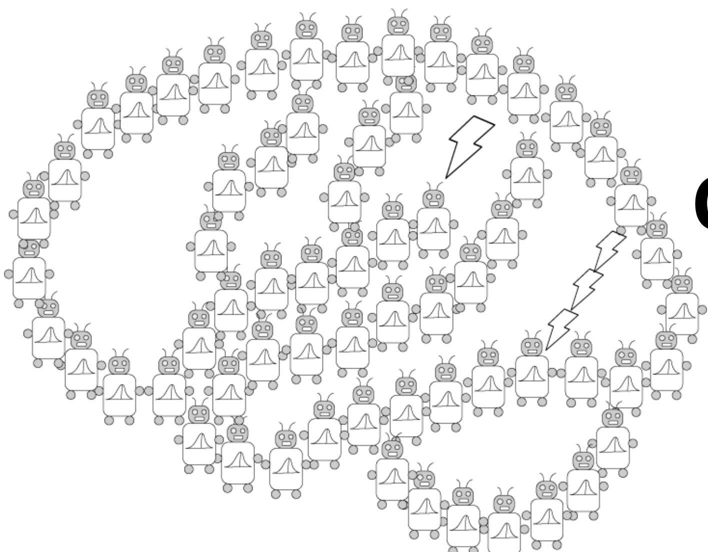
Levels of analysis and Bayes 2



PSYCH/COMP SCI/BMI 841: Computational Cognitive Science

Joseph Austerweil

January 30, 2025

- Admin Matters
- Bayes practice and thinking generatively
- Levels of analysis
- **Short rest 1**
- Math to cover:
  - Expected value
  - Continuous probability
  - Gaussian
  - Mixture modeling
  - Beta-Binomial (discrete enters the continuum)
- **Short rest 2**
- Student intros + one paper presentation
  - Go over paper presentation advice here too.

Administrative Matters: Recordings

- Class recordings getting woven into module pages in Canvas.
  - Would people prefer one page with all the links instead?
  - Doing two is challenging – alternatively could make a student writeable page for updating the lectures.

Administrative Matters: Discussion? + Quiz

- Extra discussion section
  - As of now, won't be doing it.
  - (Not feasible for this or next two weeks)
  - Will reassess Feb 17-21

- Nice job on the quiz everyone
- Q3: Apologies on the two equivalent answers
- Q4: Frequentist/Bayesian debate not as important as people being not Bayesian at all.

Administrative Matters: Next Week

- No in-person or virtual sync on Thursday 2/06.
- No OH next week. (can schedule 1-1 if needed. May need to be outside normal WI work hours)
- Please note that Japan will be 13 hours ahead
  - Email response may be more delayed than normal.
  - I'll be in meetings all day Feb 5, 8, and 9 (Japan time)
  - I will be on plane all day Feb 2 and 11 (Central time)
- I will release lecture recordings early next week
  - For class participation credit, please provide a comment/question to the associated discussion page for at least one recording.
- Paper presentations
  - Any student presentations will be recorded and posted to Canvas.
  - For class participation credit, will want you to provide one comment on at least one of them.
- First possible discussion post due 02/05 at 8:59pm
- Quiz will be released Monday 02/03 at 8am
- Quiz due back by 02/05 at 8:59pm

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Bayes' rule

Likelihood

Prior probability

Posterior probability

$$P(d|o) = \frac{P(o|d)P(d)}{\boxed{\boxed{\sum_d} P(o|d)P(d)}}$$

Total probability

Sum over diseases

Thinking generatively…

How are the hypotheses generated?

– defines the prior $p(h)$

How do the hypotheses generate the data?

– defines the likelihood $p(d|h)$

With these quantities specified, Bayes' rule inverts the generative process.

Bayes' rule

Prior probability

Likelihood

Posterior probability

Total probability

$$P(d|o) = \frac{P(o|d)P(d)}{\boxed{\boxed{\sum_d} P(o|d)P(d)}}$$

Sum over diseases

Given you hear your friend cough, what is the probability that she has...

$$P(d|o = \text{Cough})$$ for

$$d_1 = \text{Cold}$$
$$d_2 = \text{Stomach Virus}$$
$$d_3 = \text{Lung Cancer}$$

One of your friends recently became sick. Concerned about them, you go with them to the doctor. The doctor notes that your friend may have contracted **Yooper Syndrome**, which is a terrible, rare illness. Left untreated, someone with Yooper Syndrome experiences extreme pain when they are outside of the Upper Peninsula of Michigan.

1 in 10,000 people are afflicted with **Yooper Syndrome**.

To determine whether your friend has Yooper Syndrome, the doctor has your friend eat a fried cheese curd.

9 in 10 people afflicted with Yooper Syndrome foam from the mouth for 20 minutes after eating a single fried cheese curd (the extra salivation due to the extreme desire for more fried cheese curds).

Only **7 in 10 people unafflicted with Yooper Syndrome** foam from the mouth for 20 minutes after eating a single fried cheese curd.

You see your friend eat a single friend cheese curd at the doctor's office and foam from the mouth for 20 minutes.

Given this, what’s the probability that they have the Yooper?

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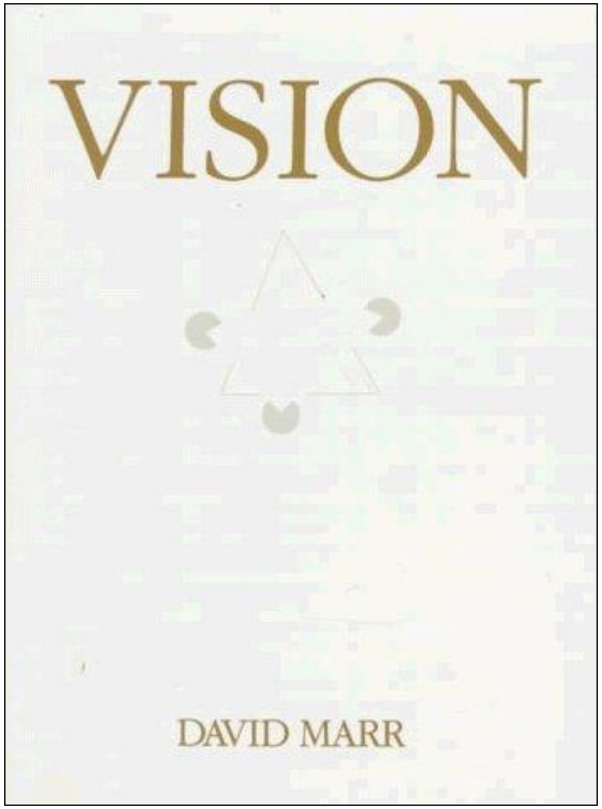
David Marr

David Marr
1945-1980



1982

Analyzing functional systems

What is being computed?

Why is it being computed?

Why is it being computed?

Justification of that formal system in terms of function

e.g. cash register: addition

- buying nothing costs nothing
- order of purchase is irrelevant
- grouping does not affect total
- Purchase + refund is zero

addition is the formal system that satisfies these functional constraints

What is being computed?

Identification of the formal system in operation

e.g. cash register: addition

- zero element $3+0 = 3$
- commutative $3+4 = 4+3$
- associative $(3+4)+5 = 3+(4+5)$
- inverses $4+(-4) = 0$

Marr (1982)'s three levels of analysis

Computation

constrains ↕

"What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?"

Representation and algorithm

constrains ↕

"What is the representation for the input and output, and the algorithm for the transformation?"

Implementation

"How can the representation and algorithm be realized physically?"

What representation?

Many formal systems solve the same computational problems.

2+2 in binary = 4 in binary

- e.g. 2+2=4 and 010+010=100

What is the mapping between representations and the inputs and outputs to the system?

Different representations make certain operations easier or more difficult

- e.g. finding powers of 10 or powers of 2

Analyzing functional systems

What is the physical implementation of the system?

Computational models can be defined at all three levels

Computational (*Normative*)

Models based on optimal solutions to abstract computational problems

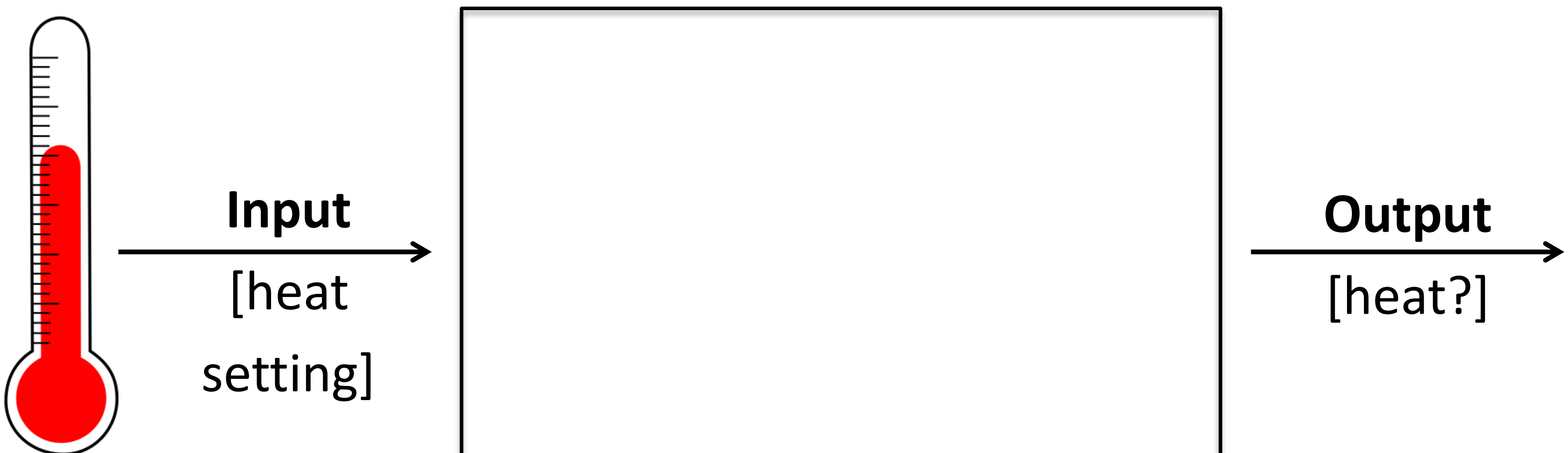
Representation and algorithm (*Descriptive*)

Models based on cognitive processes

Implementation (*Descriptive*)

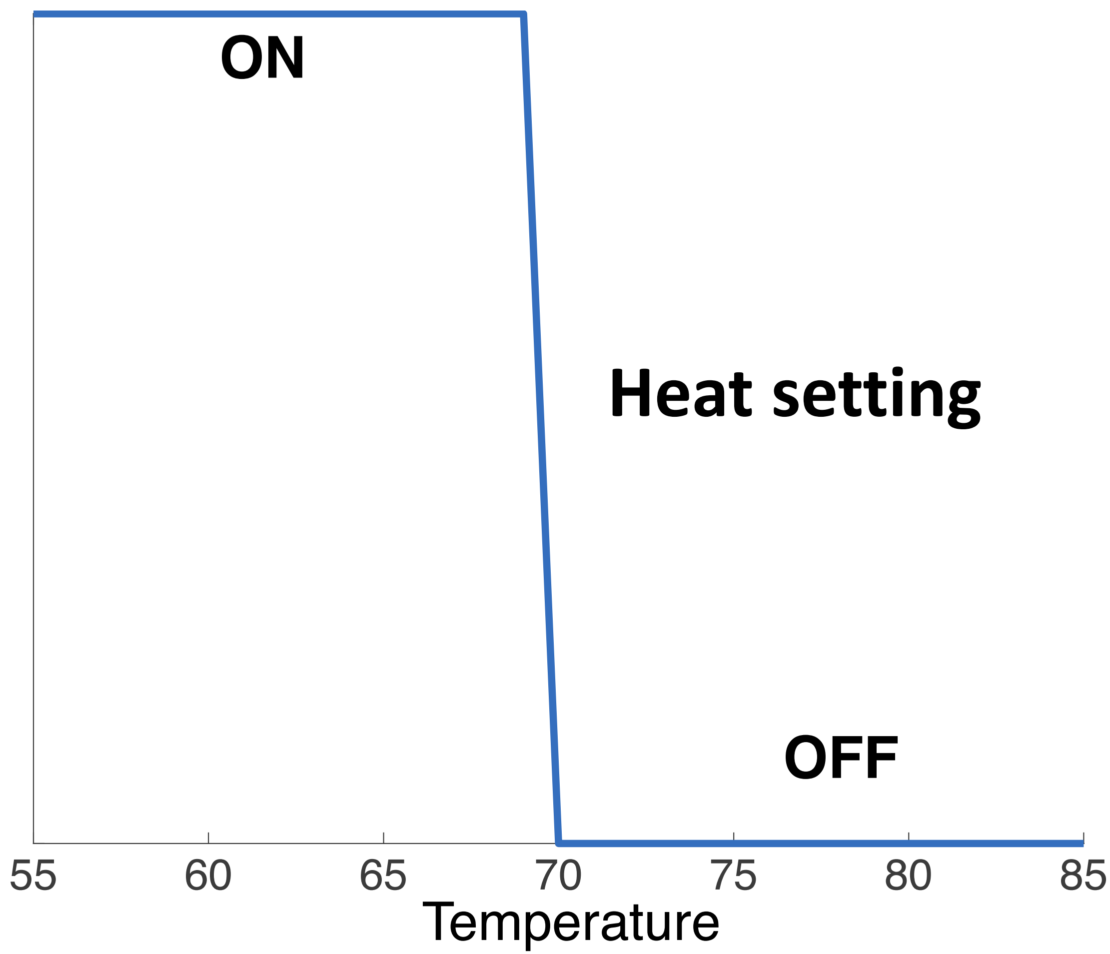
Models based on how neurons compute

The levels of a thermostat



What is the computational level description?

Computational level description



Sends ON when input temperature is at heat setting.

Algorithmic level description

1. **Read heat setting.**
2. **If temperature is less than the heat setting, then send ON to furnace.**
3. **Otherwise, do nothing.**

How does this differ from computational level description?

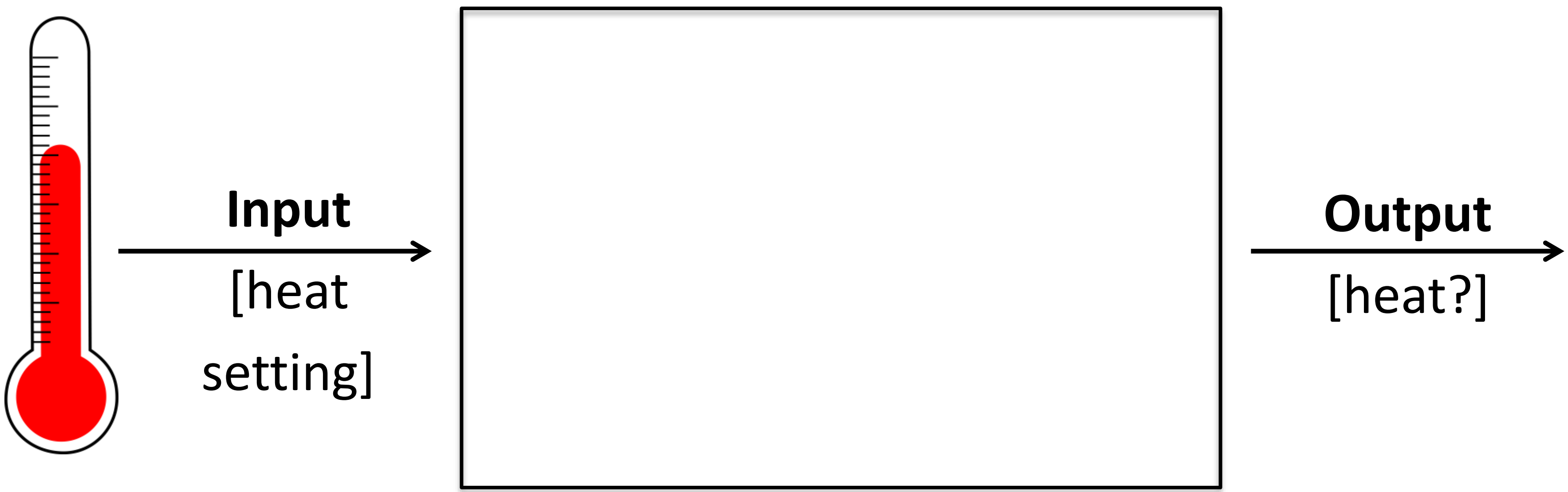
Computational level has no structure in it.*

It is just a list of ON and OFF.

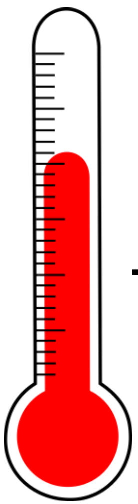
Algorithmic level describes how temperature and heat setting are compared and sends ON or OFF.

It describes a *procedure.*

Implementational level of thermostat

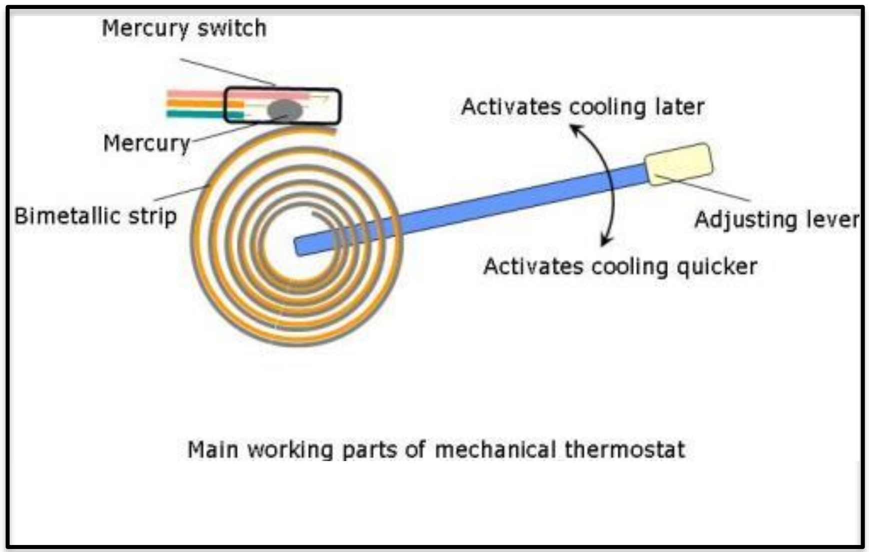


Implementational level of thermostat



Input

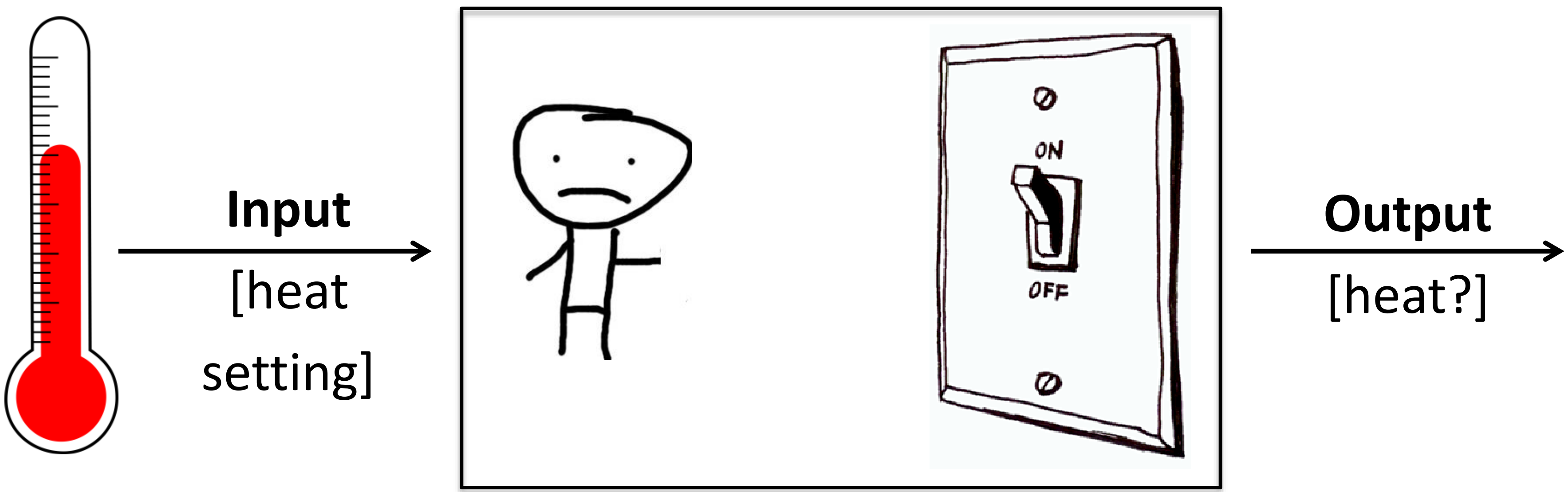
[heat setting]



Output

[heat?]

Implementational level of thermostat

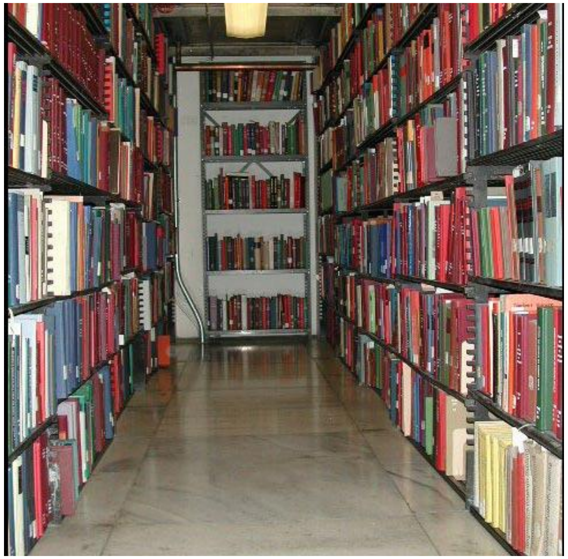


An example: Memory

An example: Memory

Computational

Explaining human memory as an optimal solution to the statistical problem of identifying items likely to be needed again



Google™

(Anderson, 1989; Griffiths, Steyvers, & Firl, 2007)

An example: Memory

Representation and algorithm

Explaining human memory based on the representation of items with binary features, and a rule for determining familiarity

Item 1: `00101011`

Item 2: `01010010`

Item 3: `11011010`

…

Old probe: `00101011`

New probe: `01111011`

Activate items with matching features

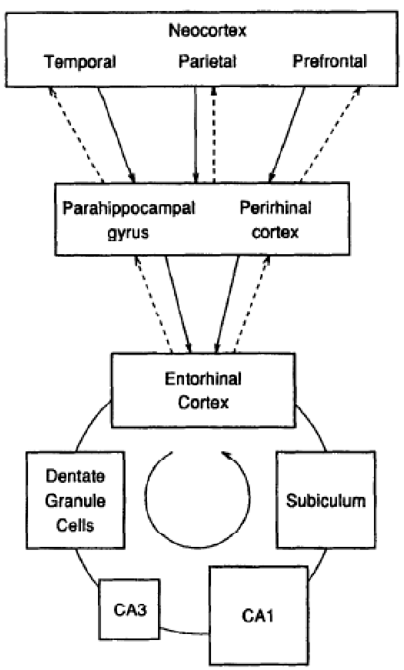
Familiarity is sum of all activation

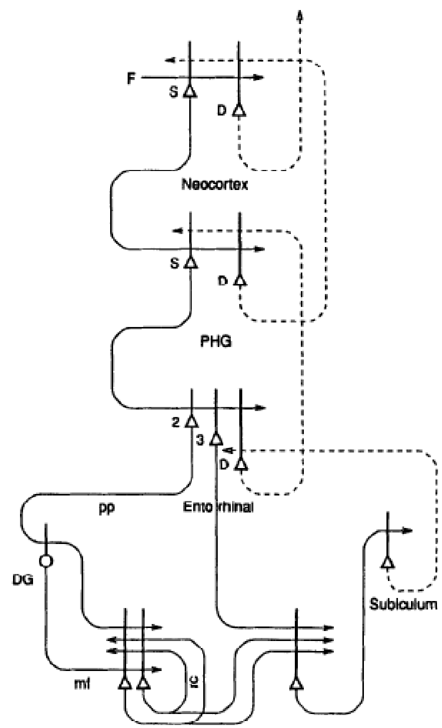
(e.g. Hintzman, 1988)

An example: Memory

Implementation

Explaining human memory based on the circuits formed by hippocampal neurons



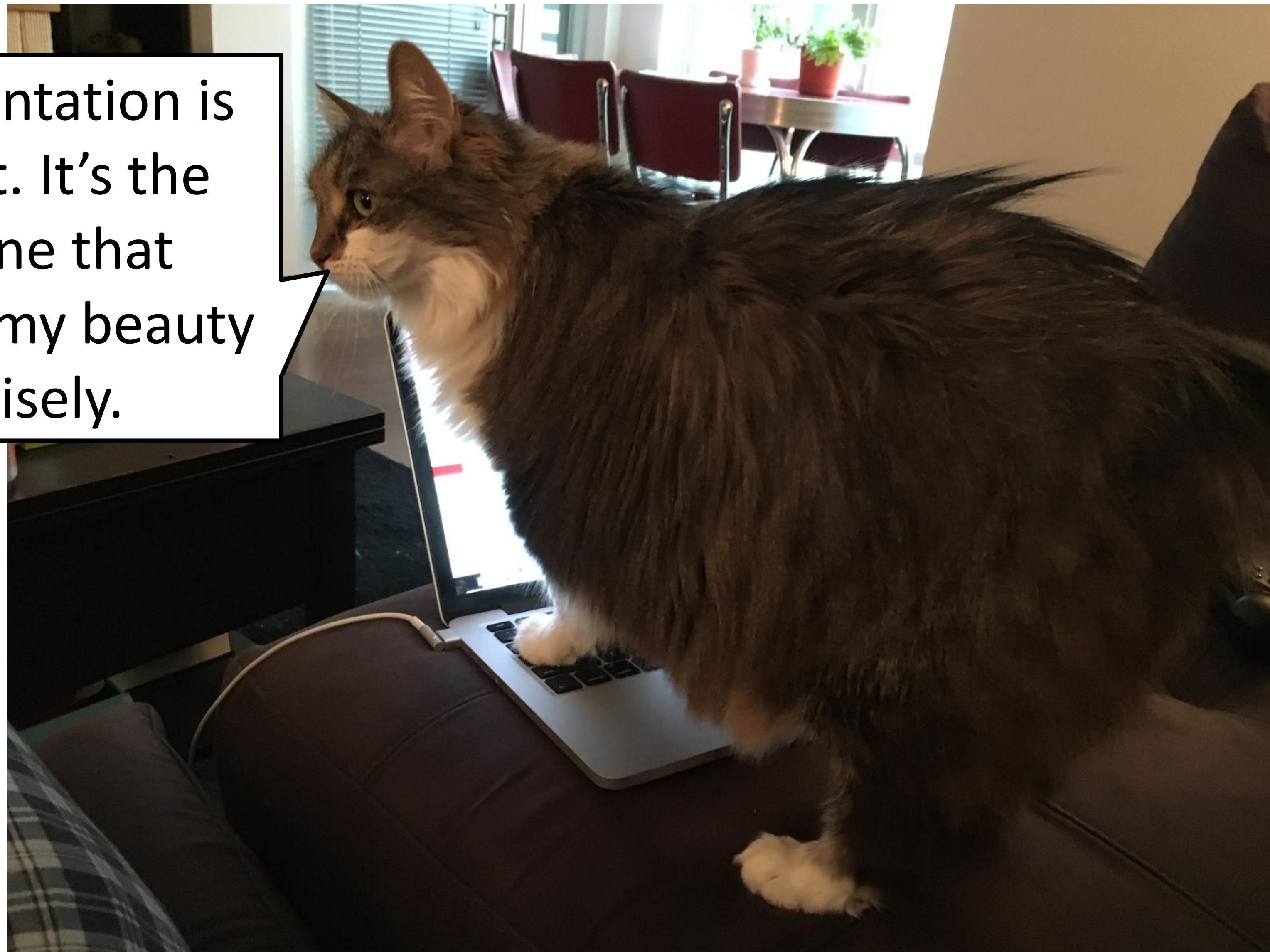


What computations are supported by different kinds of neurons?

What is the capacity of memory systems made from these neurons?

(Treves & Rolls, 1994)

Which of Marr's levels is most important?



Which of Marr’s levels is most important?

Computational/Algorithmic is the best. It captures how different things play with me. I don’t care if they’re cats, plastic, human, or feather.

Is one level more important?

Marr: the computational level is most important, imposing the most constraints

Is one level more important?

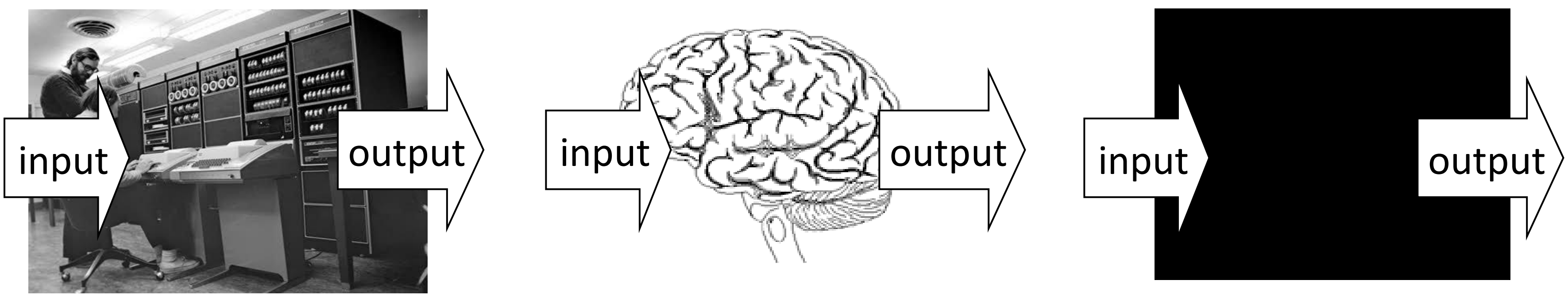
Marr: the computational level is most important, imposing the most constraints

> "...trying to understand perception by studying only neurons is like trying to understand bird flight by studying only feathers: It just cannot be done. In order to understand bird flight, we have to understand aerodynamics; only then do the structure of feathers and the different shapes of birds' wings make sense."

Only the computational level gives a *teleological* (vs. *mechanistic*) explanation

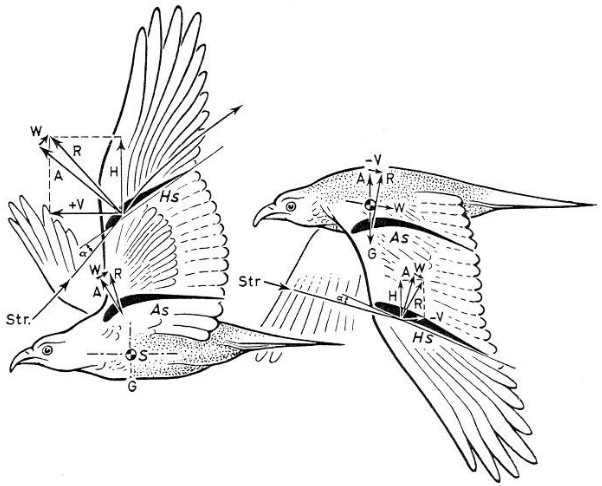
Is one level more important?

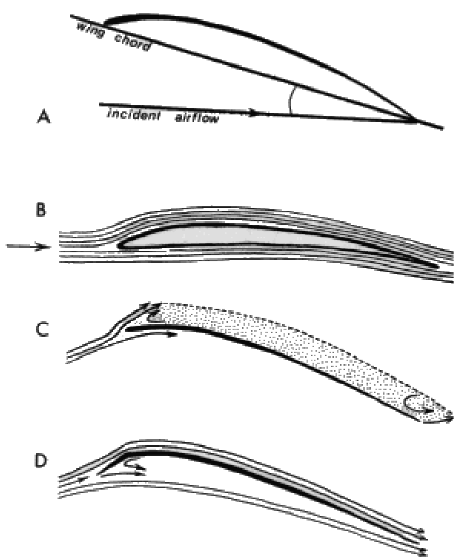
The computational level is also the one at which it doesn't matter whether we're studying humans or machines…



Provides the potential for insights to cross from one discipline to another

Different kinds of explanation





Mechanistic: how?

– algorithm
– implementation

Teleological: why?

– function/problem
– optimal solution

Bad teleological explanations…

Some properties of the structure of organisms are explained by their history, not their function

- e.g. male nipples

Reuse of existing structure (*Exaptations*)

- e.g., feathers of birds (temp. regulation)
- Are noses for wearing glasses?

Some aspects of human cognition are going to be explained by the structure of our brains and our cognitive capacities

- looking for teleological explanations everywhere will cause trouble… it's a strategy rather than a rule

There is a teleological explanation for any behavior.

Some Pros/Cons of Each Level

Computational

Pro: Designed systems, Goal-directed behavior , Emergent phenomena, Medium independent

Con: Physical history, exaptations, always some goal that can explain any phenomena

Algorithmic/Representational

Pro: Interpretable, Easy to manipulate, Medium independent,

Con: Hard to identify, Typically task-specific

Some Pros/Cons of Each Level

Algorithmic/Representational

Pro: Interpretable, Easy to manipulate, Medium independent, focuses on info. processing

Con: Hard to identify, Typically task-specific

Implementational

Pro: Closest to empirical reality, *reductive*

Con: Medium-specific, how reductive should it be?

Should we explain cognition with atoms?

Computational models can be defined at all three levels

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Bayesianism/Rational analysis

Bayesian inference is the mathematics of inductive problems.

Steps to forming a Bayesian model:

1. Formalize the problem that people face.
2. Formalize the types of knowledge people have when solving the problem.
3. Apply Bayes' rule and see what an ideal learner/observer would learn from the data.
4. How does the ideal learner behave?
5. What types of knowledge (*and/or constraints on learning) must be assumed for the model and people to behave similarly?

Explanatory virtues

(adapted from Godfrey-Smith, 2001)

In what ways can a theory be explanatory?

Empirical: The theory predicts the phenomena of interest and it is the causally relevant factor.

> *Reductionism:* Should we only work on particle mechanics then?
>
> But what about Newton's laws of physics?

Explanatory: Regardless of its empirical validity, the theory helps us understand the phenomena of interest. It is the answer to the *big* Qs of the domain.

Methodological: The theory is a useful guiding principle (heuristic) for designing experiments.

Bayesian Fundamentalism

Existence proof demonstrating the Bayesian model that exhibits the same behavior as people in some domain.

Explicitly eschews mechanisms for how Bayes is computed.

All that matters is input-output pattern corresponds to some Bayesian model.

Relation to:

- Behaviorism?
- Evolutionary Psychology

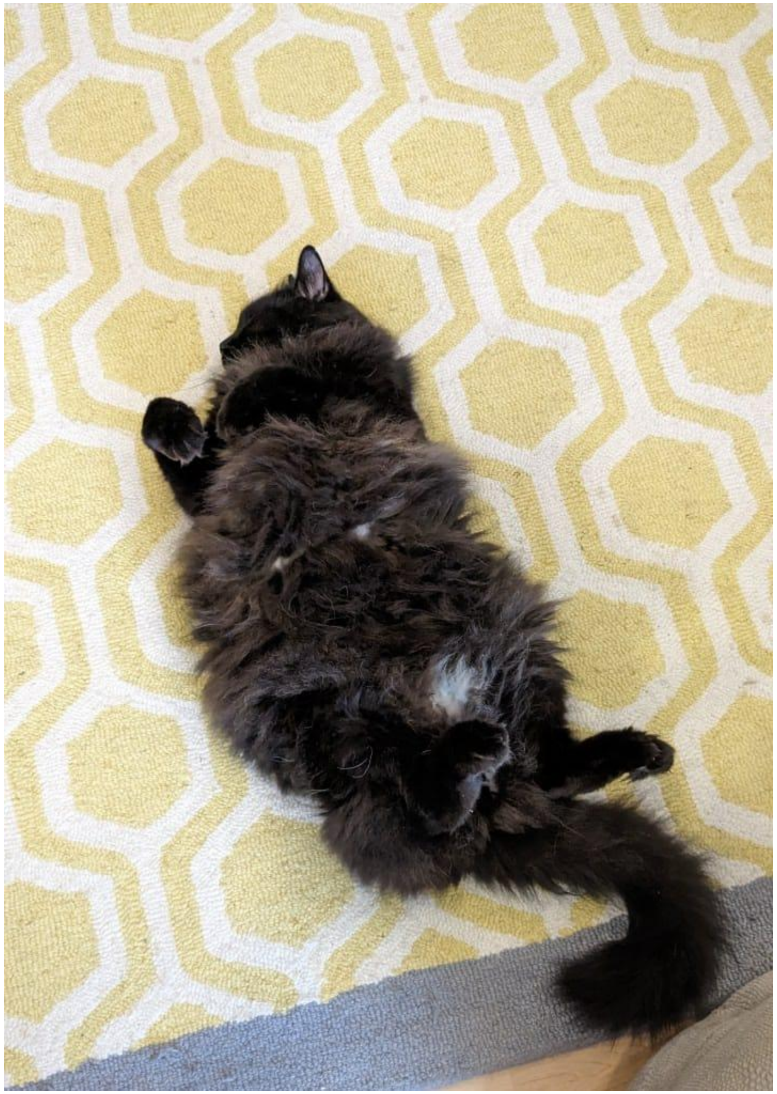
(Jones & Love, 2011)

Bayesian “Enlightenment”

1. Measure natural environment to constrain priors. (**ecological statistics**)
2. Give “psychological status” to the hypothesis space/generative models/priors.
3. Deriving “process” models from techniques used to approximate Bayesian models. Also, analyzing biases/heuristics in terms of psychological assumptions.
    1. Confirmation bias is “rational” assuming deterministic hypotheses (Austerweil & Griffiths, 2008; 2011).
    2. Analyzing neural network processes as equivalent to Bayesian models with implicit biases.
4. Integrating Bayesian models with neural mechanisms (e.g., priors as % of neurons devoted to hypothesis).

(Jones & Love, 2011)

Short rest 1



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Expected value

The expected value of a random variable (RV) X is

$$E[X] = \sum_{x} x\, P(X = x)$$

Also, called the average or mean of X.

For example, if X is a 6-sided die, the expected value is:

$$\begin{aligned} E[X] &= 1 * P(X = 1) + 2 * P(X = 1) + \cdots + 6 * P(X = 6) \\ &= 1 * \frac{1}{6} + 2 * \frac{1}{6} + \cdots + 6 * \frac{1}{6} \\ &= (1 + 2 + 3 + 4 + 5 + 6) * \frac{1}{6} = \frac{21}{6} = 3.5 \end{aligned}$$

Expected value

The expected value of a random variable (RV) X is

$$E[X] = \sum_{x} x\, P(X = x)$$

Also, called the average or mean of X.

Note:

1. Often interested in the expected value of a function $f(X)$. Since RVs are just functions, $E[f(X)] = \sum f(x) P(X = x)$.
2. **Indicator functions,** $I(A)$ (=1 if A is true, else 0), are often used to convert expectations to probabilities.
3. Expected value is linear. For RVs X and Y and constants c and d, $E[cX + dY] = cE[X] + dE[Y]$.

Expected value

The expected value of a random variable (RV) X is

$$E[X] = \sum_x x\, P(X = x)$$

For example, lets say you want to know the likely number of heads from 6 coin flips.

Let X_n be the outcome from the *n*-th coin flip.

Note*:* Heads is usually 1 and Tails is usually 0.

$$E[X_1 + X_2 + X_3 + X_4 + X_5 + X_6]$$
$$= E[X_1] + E[X_2] + E[X_3] + E[X_4] + E[X_5] + E[X_6]$$
$$= 6\, E[X_1] = 6 * (1 * 0.5 + 0 * 0.5) = 3$$

The Bernoulli Distribution

N fair coinflips $X_1, X_2, \ldots, X_N$ can be represented as

"**independent** and **identically distributed** as"

Parameter for distribution

$$X_n \overset{iid}{\sim} \text{Bernoulli}(0.5)$$

"distributed as"

"type of distribution"

The Bernoulli Distribution

N fair coinflips $X_1, X_2, \ldots, X_N$ can be represented as

$$X_n \overset{iid}{\sim} \text{Bernoulli}(\cancel{0.5}) \;\; \theta \qquad \theta \in \boxed{[0,1]}$$

Heads or 1

with probability ~~0.5~~ θ

Continuum from 0 to 1 (inclusive)

$$P(X_n) = \begin{cases} H/1 & \text{w. p.} \quad 0.5 \\ T/0/-1 & \text{o. w.} \end{cases}$$

otherwise

Tails or 0 or -1

$$E[X_n] = \sum_{a \in \{1,0\}} aP(X_n = a) = 1 \times P(X_n = 1) + 0 \times P(X_n = 0) = 1 \times 1/2 + 0 \times 1/2 = \cancel{1/2} \;\; \theta$$

What is the expected value for the sum of two coinflips?

$$E[X_1 + X_2] = E[X_1] + E[X_2] = 1/2 + 1/2 = \cancel{1} \;\; 2\theta$$

N coinflips?

$$E\left[\sum_{n=1}^{N}\right]$$

N/A

Sum from 1 to N (with index n)

Probability, Bernoulli and the Expectation Indicator Function "Trick"

The **indicator function** $I(\cdot)$ returns 1 when its argument is true and 0 otherwise.

E.g., $I(10 * 5 < 0) = 0$ or $I(\pi > 3) = 1$

$$E\left[I(X_n = 1)\right]$$

$$E\left[I(X_n = 1)\right] = \cancel{P(X_n = 1)}$$

$\Rightarrow$Convert between expectations and probabilities

$\Rightarrow$Will be critical during the approximation unit

For an event A and random variable Y,

$$P(Y \in A) = E\left[I(Y \in A)\right]$$

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Probabilities in Continuous Land

Probability mass function: $P(\cdot)$

probability mass assigned to a specific value

Discrete random variable (rv) X and continuous rv Y

$$P(X = a) > 0 \qquad\qquad \forall a^*, P(Y = a) = 0$$

Probability density function: $p(\cdot)$

Relative likelihood of a value near that precise location

$$p(y = a) \geq 0$$

To get probabilities for continuous distributions, you calculate the probability it takes a value in some **region**.

The probability is the integral of the density for a region

$$P(Y \in A) = \int_A p(y = a)\, da$$

Very small change in a

Integral symbol

Region to integrate

Probabilities in Continuous Land

Probability mass function (PMF): $P(\cdot)$

probability mass assigned to a specific value

Discrete random variable (rv) X and continuous rv Y

$$P(X = a) > 0 \qquad\qquad \forall a, P(Y = a) = 0$$

Probability density function (PDF):

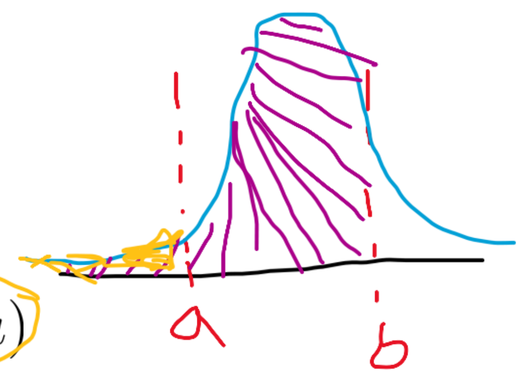
Relative likelihood of a value near that precise location

$$p(y = a) \geq 0$$

Note: $$\boxed{P(Y \in \Omega) = \int_{\Omega} p(y = a)da = 1}$$

Cumulative distribution function (CDF)

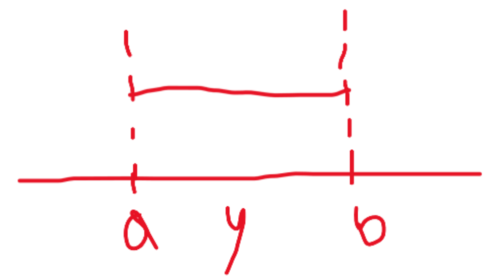
$$F(y) = \int_{-\infty}^{y} p(a)da \qquad P(Y \in [a, b]) = F(b) - F(a)$$



Continuous Uniform Distribution

The continuous uniform distribution over the interval $[a,b]$ is

$$p(Y = y) = \frac{1}{b-a} I(y \in [a, b])$$



$$Y \sim \text{Uniform}\,([a, b])$$

"distributed as"

E.g., $Y \sim \text{Uniform}\,([0, 0.5])$ $\qquad p(Y = 0.25) = \dfrac{1}{0.5 - 0} = 2$

What does that mean?

How about prob. that Y is less than 0.1?

$$P(Y \in [0, 0.1]) = \int_0^{0.1} 2dy = 2 \times 0.1 = 0.2$$

Expected value? $E[Y] = \int yp(y)dy$

$$E[Y] = \int_0^{0.5} yp(y)dy =$$

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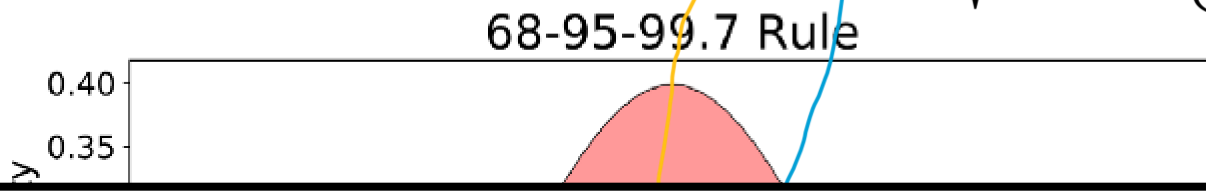
Gaussian/Normal Distribution

$$Y \sim \text{Gaussian}(\mu, \sigma^2)$$

Alternative parametrization: $\tau = 1/\sigma^2$ $\quad p(y|\mu,\tau) = \sqrt{\frac{\tau}{2\pi}} \exp\left\{-\frac{\tau^2}{2}(y-\mu)^2\right\}$



Note: There is no closed form solution for the CDF.

A special case of the CDF is the "Error Function".

It is often written as Φ.

$$p(y|\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2}(y-\mu)^2\right\}$$

Image from https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2

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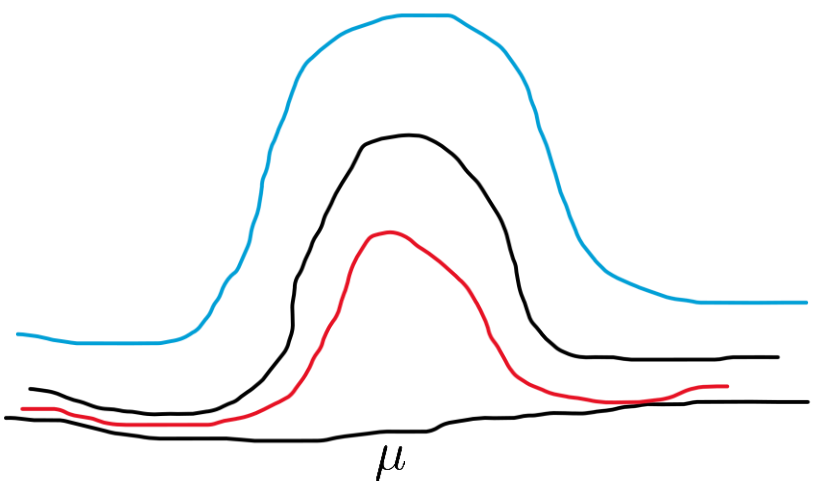
Posterior Mean For Normal Prior Given Known Variances

$$\mu \sim \text{Gaussian}(\mu_0, \sigma_0^2)$$

$$Y|\mu \sim \text{Gaussian}(\mu, \sigma_Y^2)$$

$$p(\mu|Y) = \frac{p(Y|\mu)p(\mu)}{p(Y)}$$

for some $k > 0$ (k does not depend on μ)



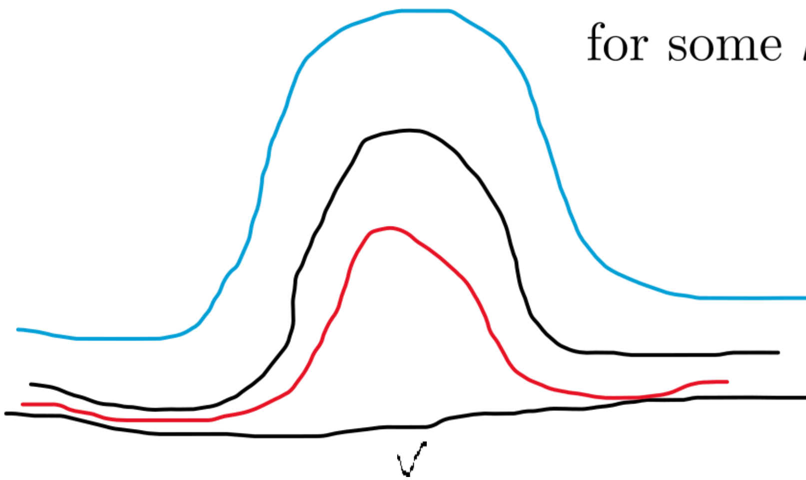
Varying k just stretches the function vertically – neither changes the functional form nor ordering of which points have greater values than others.

k will be the value that makes the integral of the function 1 for the given Y. WHY?

Posterior Mean For Normal Prior Given Known Variances

$$p(\mu|Y) = \frac{p(Y|\mu)p(\mu)}{p(Y)} = kp(Y|\mu)p(\mu)$$

for some $k > 0$ (k does not depend on μ)



Varying k just stretches the function vertically – neither changes the functional form nor ordering of which points have greater values than others.

k will be the value that makes the integral of the function 1 for the given Y. WHY?

Remember $\int kp(y|\mu)p(\mu)d\mu = 1.$

So, $\int kp(y|\mu)p(\mu) = 1$

$1/p(Y)$

“normalizing constant” or “partition function”

Posterior Mean For Normal Prior Given Known Variances

$$p(\mu|Y) = \frac{p(Y|\mu)p(\mu)}{p(Y)} = kp(Y|\mu)p(\mu) \propto p(Y|\mu)p(\mu)$$

$$= \frac{1}{\sigma_Y\sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma_Y^2}\left(Y-\mu\right)^2\right\}$$

$$= \frac{1}{2\pi\sigma_Y\sigma_0} \exp\left\{-\frac{1}{2\sigma_Y^2}\left(Y-\mu\right)^2 \qquad\qquad\qquad\qquad \right\}$$

$$\propto \exp\left\{-\frac{1}{2\sigma_Y^2}\left(Y^2-2\mu Y+\mu^2\right) \qquad\qquad\qquad\qquad \right\}$$

$$\propto \exp\left\{-\mu^2 \boxed{\left(\frac{1}{2\sigma_Y^2}+\frac{1}{2\sigma_0^2}\right)} + 2\mu\left(\frac{Y}{2\sigma_Y^2}+\frac{\mu_0}{2\sigma_0^2}\right)\right\}$$

$$\exp\left\{-\boxed{\frac{1}{2\sigma_1^2}}\left(\mu-\mu_1\right)^2\right\}$$

Posterior Mean For Normal Prior Given Known Variances

$$p(\mu|Y) = \frac{p(Y|\mu)p(\mu)}{p(Y)} = kp(Y|\mu)p(\mu) \propto p(Y|\mu)p(\mu)$$

$$\propto \exp\left\{-\mu^2 \boxed{\left(\frac{1}{2\sigma_Y^2} + \frac{1}{2\sigma_0^2}\right)} + 2\mu\left(\frac{Y}{2\sigma_Y^2} + \frac{\mu_0}{2\sigma_0^2}\right)\right\}$$

$$\exp\left\{-\frac{1}{2\sigma_1^2}(\mu - \mu_1)^2\right\}$$

$$\sigma_1^2 = \left(\frac{1}{\sigma_x^2} + \frac{1}{\sigma_0^2}\right)^{-1}$$

$$-\frac{1}{2\sigma_1^2}\left(\mu^2 - 2\mu\mu_1 + \mu_1^2\right)$$

$$\mu_1 = \cancel{\frac{1}{2}}\left(\frac{1}{\sigma_x^2} + \frac{1}{\sigma_0^2}\right)^{-1}\left(\frac{x}{\sigma_x^2} + \frac{\mu_0}{\sigma_0^2}\right)$$

Cluster assignment part 1 walkthrough

https://canvas.wisc.edu/courses/446937/assignments/2579703

- Admin Matters
- Bayes practice and thinking generatively
- Levels of analysis
- **Short rest 1**
- Math to cover:
  - Expected value
  - Continuous probability
  - Gaussian
  - Mixture modeling
  - Beta-Binomial (discrete enters the continuum)
- **Short rest 2**
- Student intros + one paper presentation
  - Go over paper presentation advice here too.

Discrete Distribution

Prof. Joe is walking through an animal shelter.

There are three animals there: ferrets, cats, and dogs.

How do we model the probability that he encounters a ferret, cat or dog?

We can model this with $\vec{\theta} = (\theta_1, \theta_2, \theta_3)$, one for each animal for $(\sum_k \theta_k = 1)$.

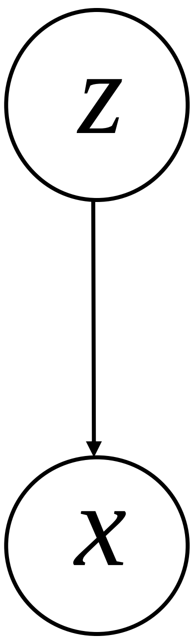
We can define a random variable c to be the next animal encounter. It is **Discrete** distributed:

$$c|\vec{\theta} \sim \text{Discrete}(\vec{\theta})$$

It is sometimes called a *categorical* distribution.

$$P(c = k) = \theta_k$$

Mixture models



$P(z)$ Latent (unobserved)

$P(x|z)$ Observed

$$P(x) = \sum_z P(x \mid z) P(z)$$

Marginalize out
(i.e. sum over)
latent structure

Plate notation

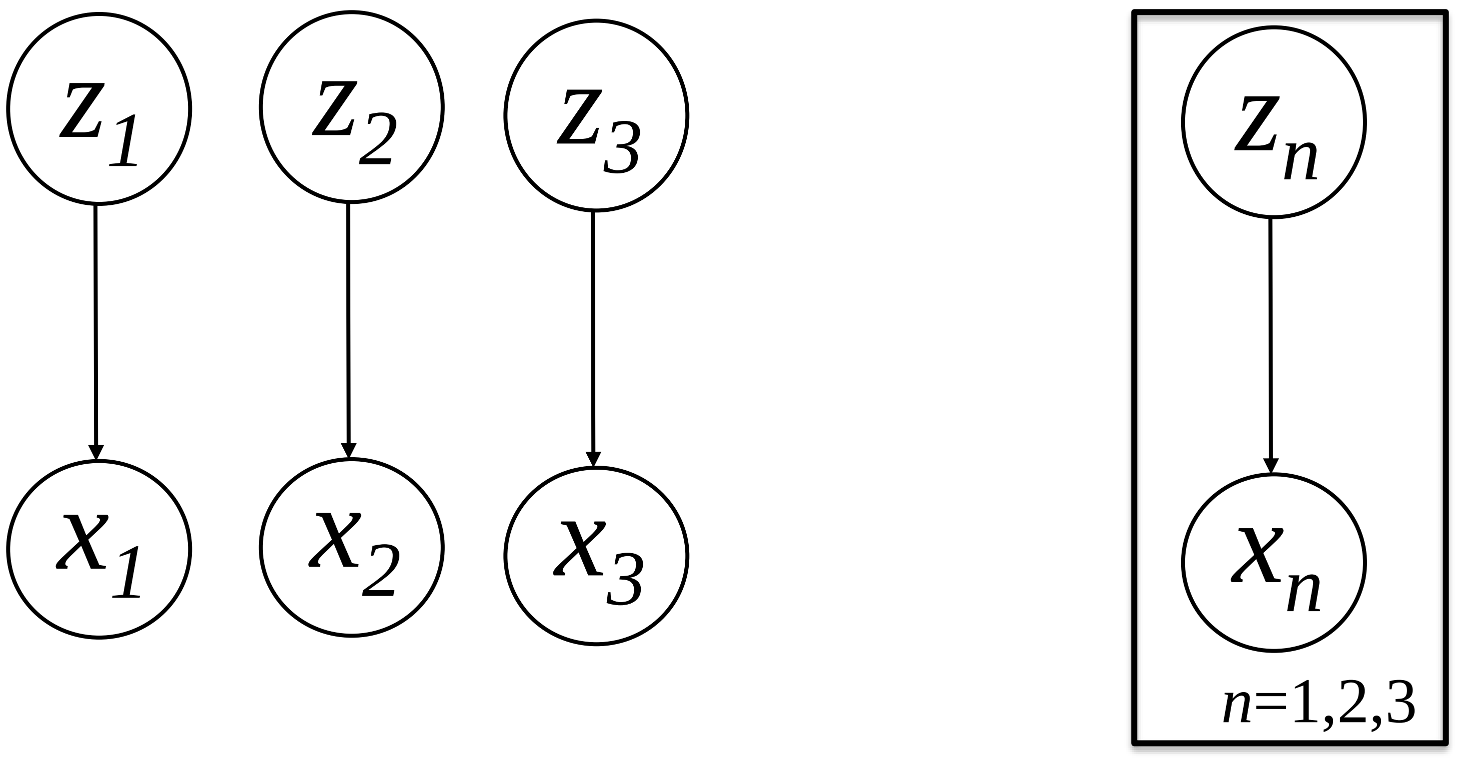
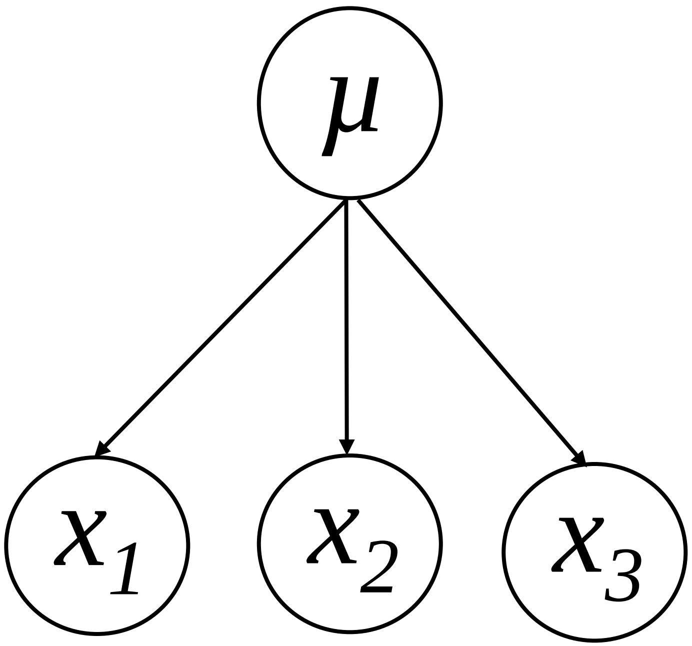
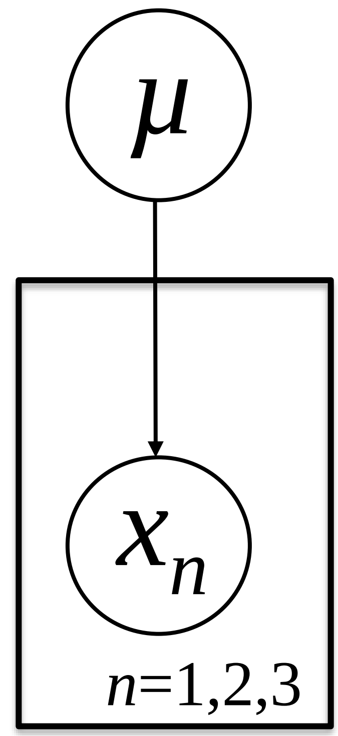


Plate notation





Categorization as Mixture Modeling

To make an animal, first sample which animal we want from $\theta = (.2, \quad .4, \quad .4)$

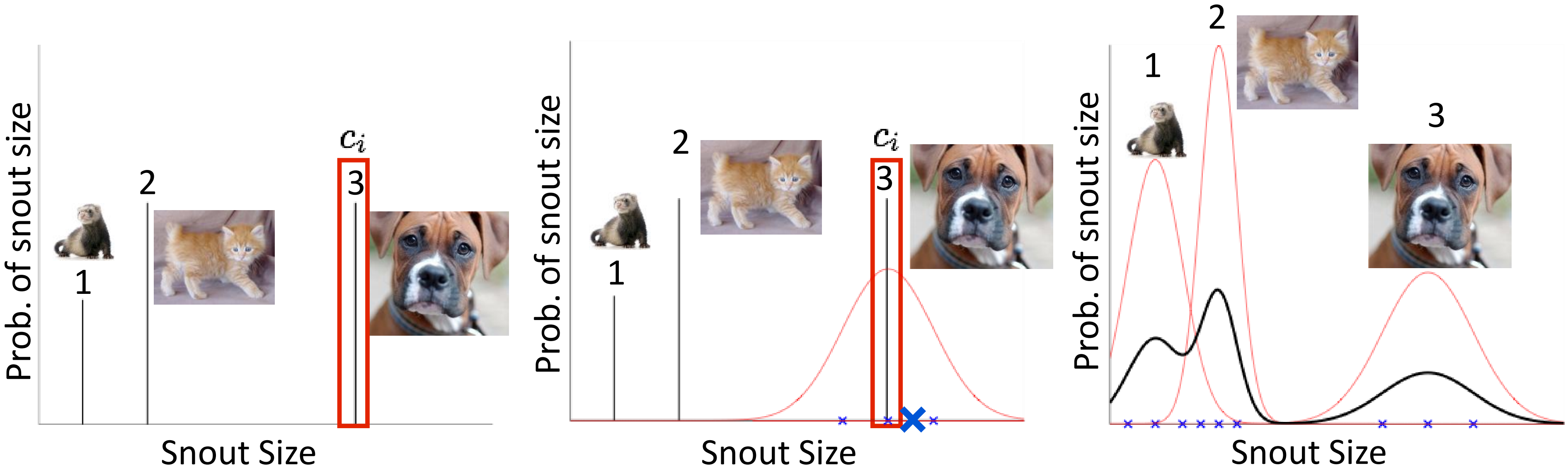
Each animal is created from its associated ideal “snout size” μ_c and variance σ_c^2.

This leads to the following *generative process* for each animal

$$\boxed{c_i|\theta \sim \text{Discrete}(\theta)} \qquad \boxed{x_i|c_i \sim \text{Gaussian}(\mu_{c_i}, \sigma_{c_i}^2)}$$

Prob. of new snout size: $P(x_i) = \sum_{k=1}^{3} P(x_i, c_i = k) = \sum_{k=1}^{3} \boxed{P(x_i|\mu_k, \sigma_k^2)}\boxed{P(c_i = k|\theta)}$

Categorize a new animal given its snout size: $P(c_i = 1|x_i, \theta) = \dfrac{p(x_i|\mu_1, \sigma_1^2)P(c = 1|\theta)}{\sum_{k=1}^{3} p(x_i|\mu_k, \sigma_k^2)P(c = k|\theta)}$



Cluster assignment part 2 walkthrough

- https://canvas.wisc.edu/courses/446937/assignments/2579703

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Binomial Distribution

$$X_1, \ldots, X_N | \theta \overset{iid}{\sim} \text{Bernoulli}(\theta) \quad Y = \sum_{n=1}^{N} X_n \quad \Rightarrow Y \sim \text{Binomial}(N, \theta)$$

# of heads from N coinflips.

Create N - k tails

$$P(Y = k|\theta) = \binom{N}{k} \theta^k (1-\theta)^{N-k}$$

Create k heads

Count the number of ways you can get k heads with N coinflips. "N choose k"

Number of ways to order N unique items

$$\binom{N}{k} = \frac{N!}{k!(N-k)!}$$

Number of ways to order N - k tails

Number of ways to order k heads

N! is the "factorial" operator. It counts the number of ways to order N items.

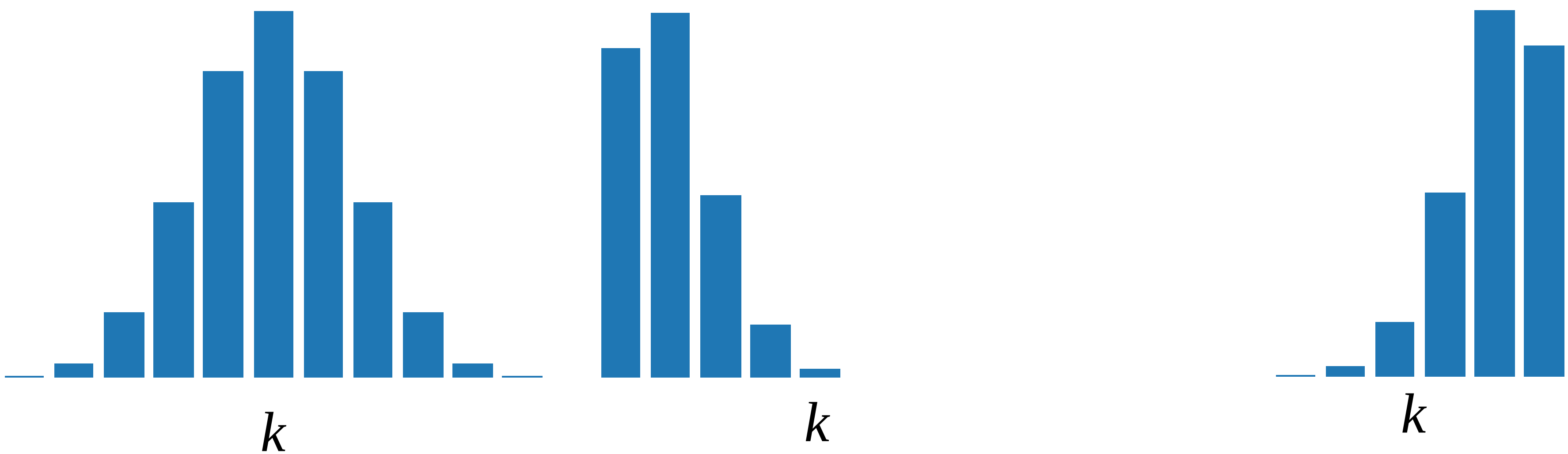
$$N! = 1 \times 2 \times \ldots \times N = \prod_{n=1}^{N} n$$

Binomial Distribution

$$X_1, \ldots, X_N | \theta \overset{iid}{\sim} \text{Bernoulli}(\theta) \quad Y = \sum_{n=1}^{N} X_n \;\; \Rightarrow Y \sim \text{Binomial}(N, \theta)$$

$$P(Y = k|\theta) = \binom{N}{k} \theta^k (1-\theta)^{N-k}$$

$$P(Y = k|\theta = 0.5) \qquad P(Y = k|\theta = 0.1) \qquad P(Y = k|\theta = 0.9)$$



What if we don't know the coinflip bias? (the probability of heads θ)?

The Beta Distribution: A distribution over the probability of a heads

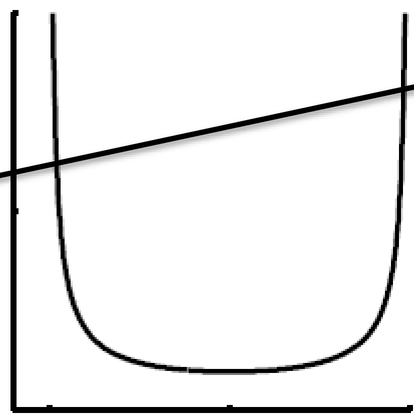
Beta(a, b)

$$p(\theta|a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1}$$

These are Beta(αβ, α(1-β)).

$$p\left(\theta|\alpha\beta,\alpha(1-\beta)\right) = \frac{\Gamma(\alpha)}{\Gamma(\alpha\beta)\Gamma\left(\alpha(1-\beta)\right)}\theta^{\alpha\beta-1}(1-\theta)^{\alpha(1-\beta)-1}$$

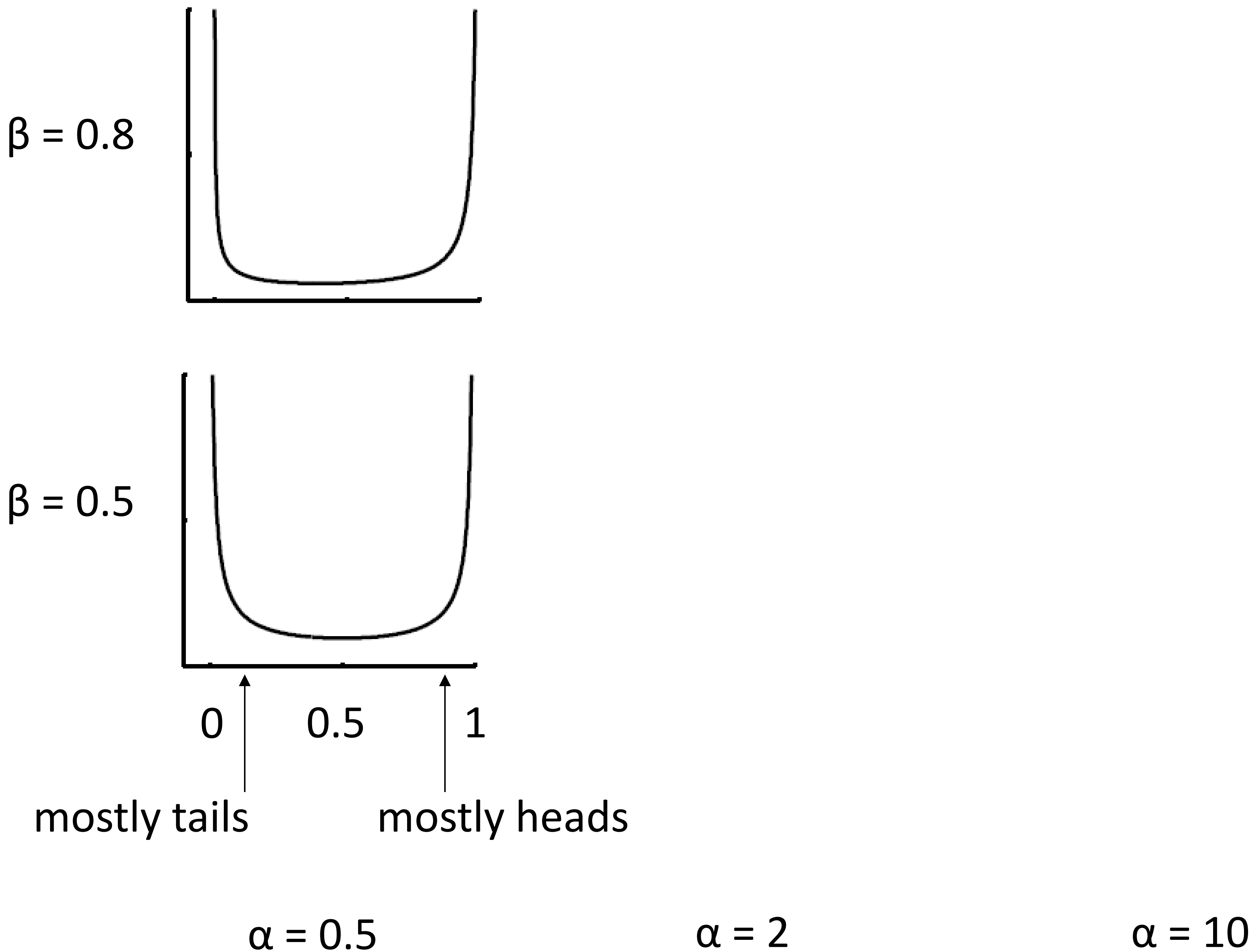
β = 0.5 → Average Value



0 0.5 1

mostly tails mostly heads

α = 0.5 → Shape


β = 0.8
β = 0.5
0
0.5
1
mostly tails
mostly heads
α = 0.5
α = 2
α = 10

Gamma function interlude

In all likelihood, never will need to deal with directly.

It is the generalized factorial function. $\Gamma(x) = \int_0^\infty z^{x-1} e^{-z} dz$

For positive integer *x*, $\Gamma(x) = (x-1)!$

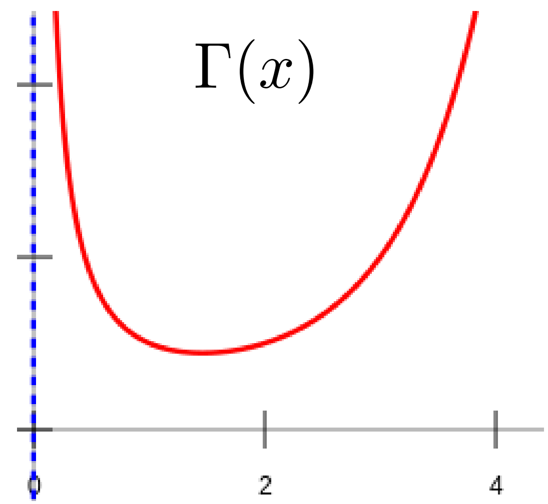
As far as I'm concerned, it is restricted to positive real numbers.

It is the normalization constant for the Gamma distribution.

Some useful facts:

$$\Gamma(x+1) = x\Gamma(x)$$

$$\Gamma(1) = 0! = 1 = \Gamma(2) = 1! = 1$$



Beta-Binomial Posterior Update

$$\theta \sim \text{Beta}(a, b) \qquad Y|\theta \sim \text{Binomial}(N, \theta)$$

$$p(\theta|a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1} \qquad P(Y=k|\theta) = \binom{N}{k}\theta^k(1-\theta)^{N-k}$$

$$p(\theta|Y=N_H) \propto P(Y=N_H|\theta)p(\theta|a,b)$$

$$p(\theta|Y=N_H) \propto \cancel{\frac{N!}{N_H!(N-N_H)!}}\theta^{N_H}(1-\theta)^{N-N_H}\cancel{\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}}\theta^{a-1}(1-\theta)^{b-1}$$

$$p(\theta|Y=N_H) \propto \theta^{N_H}(1-\theta)^{N-N_H}\theta^{a-1}(1-\theta)^{b-1} = \theta^{N_H+a-1}(1-\theta)^{N-N_H+b-1}$$

$$\Rightarrow \theta|Y=N_H \sim \text{Beta}(N_H+a, N-N_H+b)$$

Posterior is of the same functional form as prior (Both are prior).
Parameters are updated to reflect knowledge.
New *a* adds in the number of heads. New *b* adds in number of tails.

This is a **Conjugate Prior**. (more on this next week)

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Short rest 2

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Class presentations (2x 5% each)

Present the major results of one of the course readings.

- Must be a non-optional reading.

Also, have 4-6 discussion questions at the end.

- These should guide discussion.
- Being controversial/provocative is encouraged!

Have to give two if you are taking the course for credit.

- One if formally auditing.

Each presentation (including discussion) should be no longer than 25 minutes.

You are encouraged to meet with me beforehand.
Must submit to me on Canvas right before or after (8:59pm day of presentation deadline)!

Class presentations (2x 5% each)

Grade is determined by the following factors:

Understanding of the paper	(2 pts)
Covering the key aspects of the paper	(1 pt)
Presentation clarity	(1 pt)
Appropriate discussion questions	(0.5 pts)
Appropriate use of time	(0.5 pts)
Not using Prezi	(10^{999} deduction and have to clean up)

See guidelines and tips document posted online.

Class presentations (2x 5% each)

Focus on the following questions:

1. What is the question that this paper explores? If applicable, what aspect of human behavior are we trying to understand?
2. Any background/context for the paper? Previous work?
3. What is the underlying computational problem being solved? How does the paper formulate it?
4. What is the solution to the computational problem? (Go for intuitions more than rigor.)
5. How is the solution used to answer the question? How does it compare to other approaches/human behavior?
6. What conclusions can we draw from the results? What is your interpretation?

Student introductions

1. Name, preferred pronouns, undergrad/grad, department
2. Why are you taking this course?
3. What is one of your favorite comfort foods?

- Admin Matters
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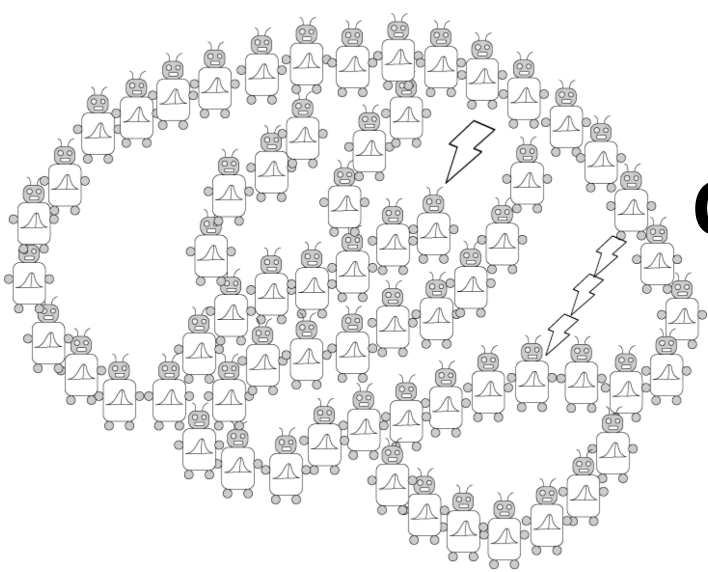
Randomness and Coincidences: Reconciling Intuition and Probability Theory (2001)

by Griffiths & Tenenbaum



PSYCH/COMP SCI/BMI 841: Computational Cognitive Science

Joseph Austerweil

January 30, 2025

Which sequence feels more random?

HHHHHHHH

or

HHTHTTTH

Background

People's perception of randomness deviates from a traditional take from probability theory.

HHTHTTTH feels more random than HHHHHHHH

People tend to perceive sequences that alternate 60% of the time as the most random (Lopes, 1982).

Theories to explain the discrepancy:

1. Different normative theory for randomness
2. *Representativeness:* Sequence similar to the output of a random generating device

Question

What normative theory should be used for randomness and do people deviate from it systematically?

Bayesianism

Bayesian inference is the mathematics of inductive problems.*

Steps to forming a Bayesian model:

1. Formalize the problem that people face.
2. Formalize the types of knowledge people have when solving the problem.
3. Apply Bayes' rule and see what an ideal learner/observer would learn from the data.
4. How does the ideal learner behave?
5. What types of knowledge (*and/or constraints on learning) must be assumed for the model and people to behave similarly?

Computational formalization

Evaluate which of two generating processes is more probable given the observation *x*:

$$\text{random}(x) = \frac{P(\text{random}|x)}{P(\text{regular}|x)}$$

Use odds form of Bayes' rule to avoid calculating normalizing constant!

$$\frac{P(\text{random}|x)}{P(\text{regular}|x)} = \frac{\frac{P(x|\text{random})P(\text{random})}{\cancel{P(x)}}}{\frac{P(x|\text{regular})P(\text{regular})}{\cancel{P(x)}}}$$

Assume equal prior odds

$$\text{random}(x) = \log\left(\frac{P(x|\text{random})}{P(x|\text{regular})}\right)$$

$$P(x|\text{random}) = \left(\frac{1}{2}\right)^N = 2^{-N}$$

What should $P(\text{regular}|x)$ be?

Computational formalization

Assume $\mathcal{H}$ is a set of possible regularities that might produce a non-random sequence.

$$P(x|\text{regular}) =$$

What if regular means a coinflip bias that is uniform over possible biases?

$$\mathcal{H} = [0, 1] \qquad\qquad h|\text{regular} \sim \mathrm{Beta}(1, 1)$$

$$P(H|\text{regular}) = \int_0^1 \underbrace{\theta^H (1-\theta)^{N-H}}_{\mathrm{Beta}(H,\, N-H)} \times 1 \times d\theta$$

So, the integral must be the normalizing constant for a $\mathrm{Beta}(H+1, N-H+1)$ distribution.

$$\Rightarrow P(H|\text{regular}) =$$

$$P(H|\text{regular}) = \frac{H!(N-H)!}{(N+1)!} =$$

Computational formalization

$$\text{random}(x) = \log\left(\frac{P(x|\text{random})}{P(x|\text{regular})}\right) \qquad P(x|\text{random}) = \left(\frac{1}{2}\right)^N = 2^{-N}$$

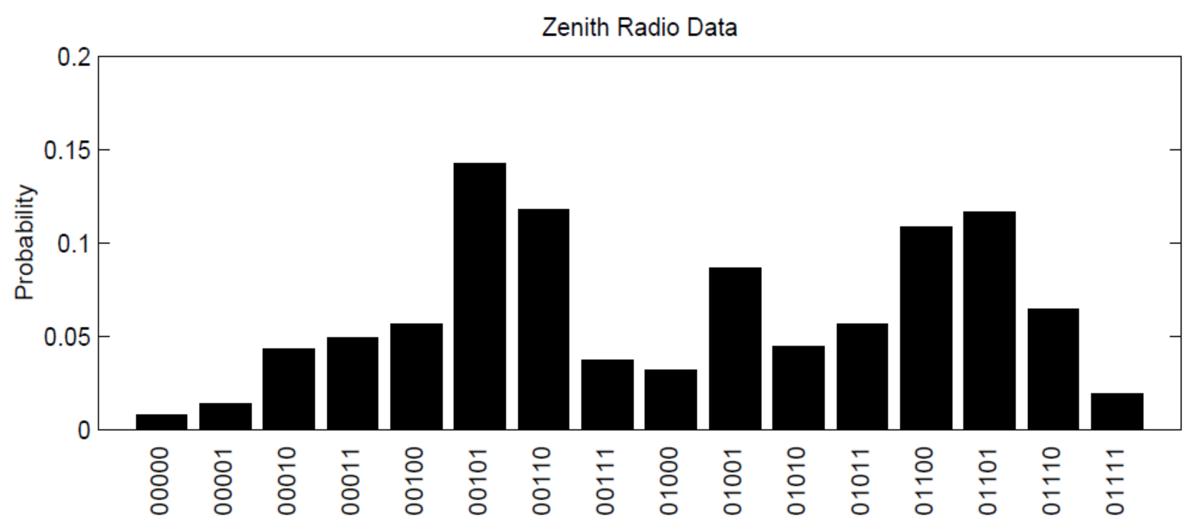
$$P(H|\text{regular}) = (N+1)^{-1}\binom{N}{H}^{-1}$$

$$\text{random}(x) = \underbrace{\log 2^{-N} + \log(N+1)}_{\text{Does not vary for sequences of the same length...}} + \log\binom{N}{H}$$

Does not vary for sequences of the same length...

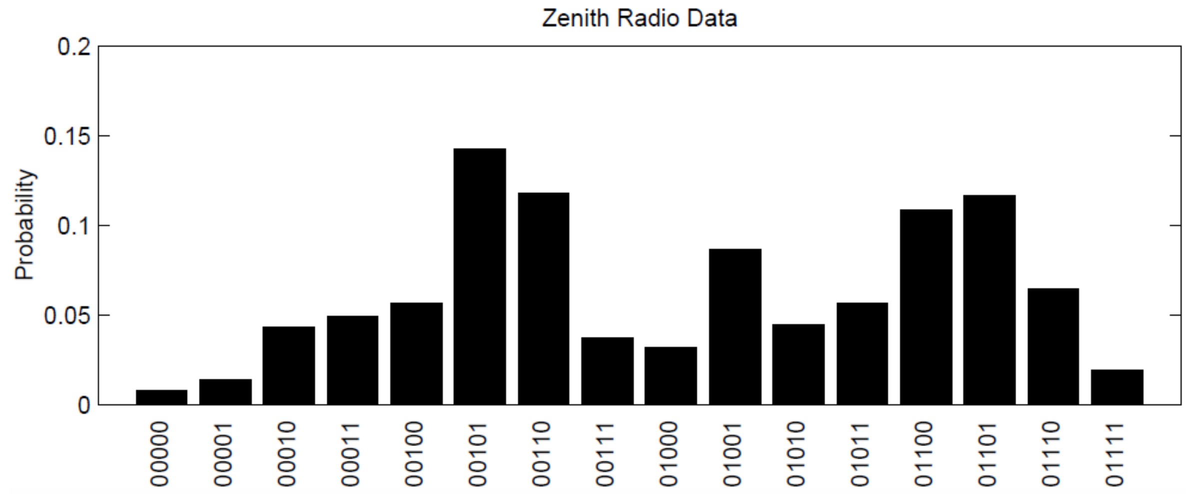
Zenith ESP "Experiment"

- In 1937, Zenith Corp. broadcasted psychics who transmitted (silently) a sequence of coinflips to listeners.
- Listeners sent in what they got from the psychics.
- Regularities, but unrelated to the psychics…



Goodfellow (1938)

Modeling the Zenith ESP "Experiment"



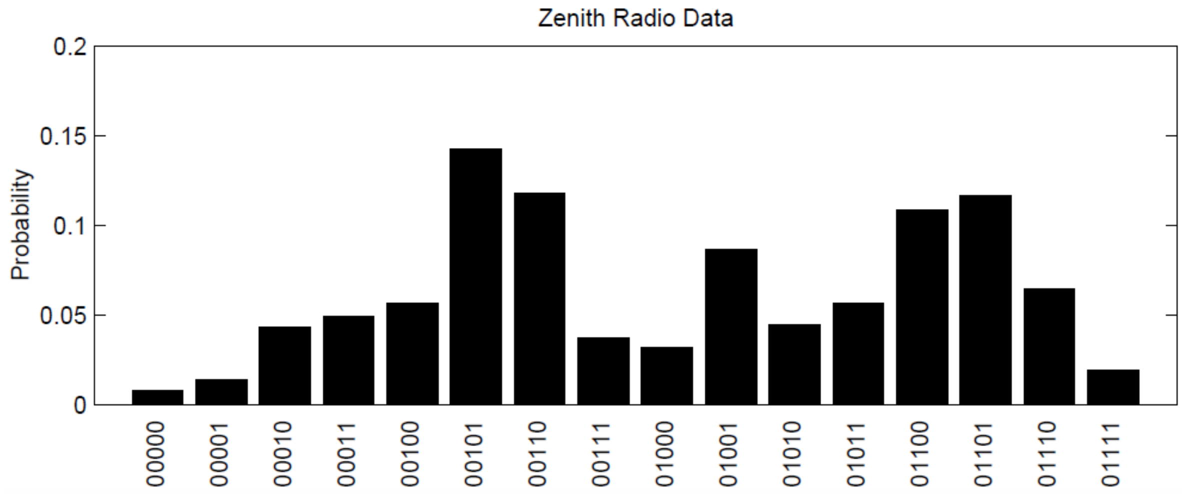
Sequentially generated, so Griffiths and Tenenbaum (2000) assume people select H vs. T as the *k*-th response based on how much more "random" they were when compared to all shorter subsequences.

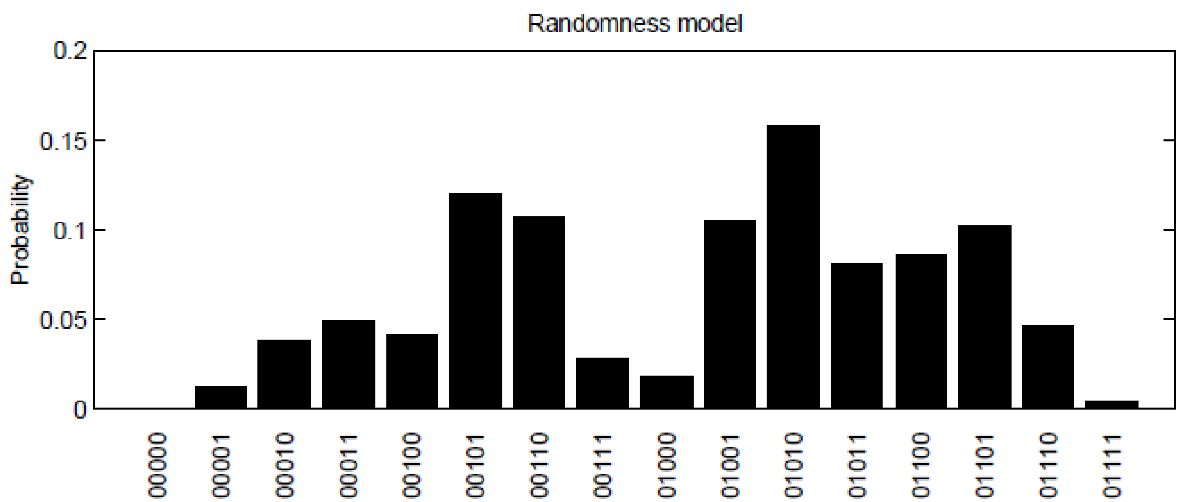
$$L_k = \sum_{i=1}^{k-1} \mathrm{random}(H_i + 1, T_i) - \mathrm{random}(H_i, T_i + 1)$$

Response then was mapped through a logistic function to allow for some uniformity over sequences in model behavior.

$$P(R_k = H) = \frac{1}{1 + e^{-\lambda L_k}}$$

Modeling the Zenith ESP "Experiment"

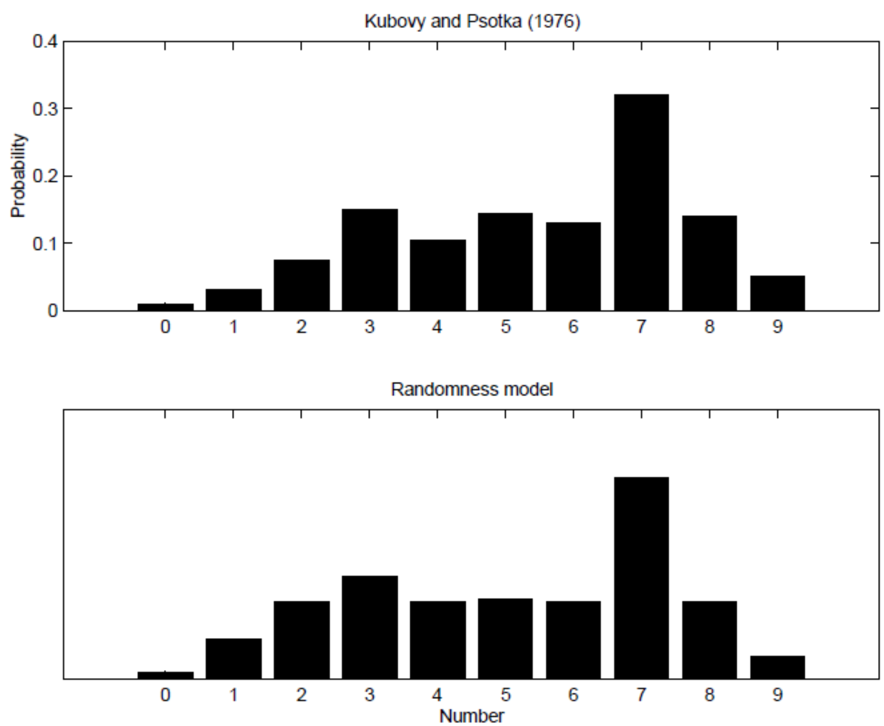




Generate digit 0 to 9

Random likelihood is uniform over 0 to 9.

Model is very similar, except now we use number properties of digits from 0 to 9 as our hypothesis space (e.g., multiples of 2, multiples of 3) and then a digit is generated uniformly at random from the digits consistent with the sampled number property.



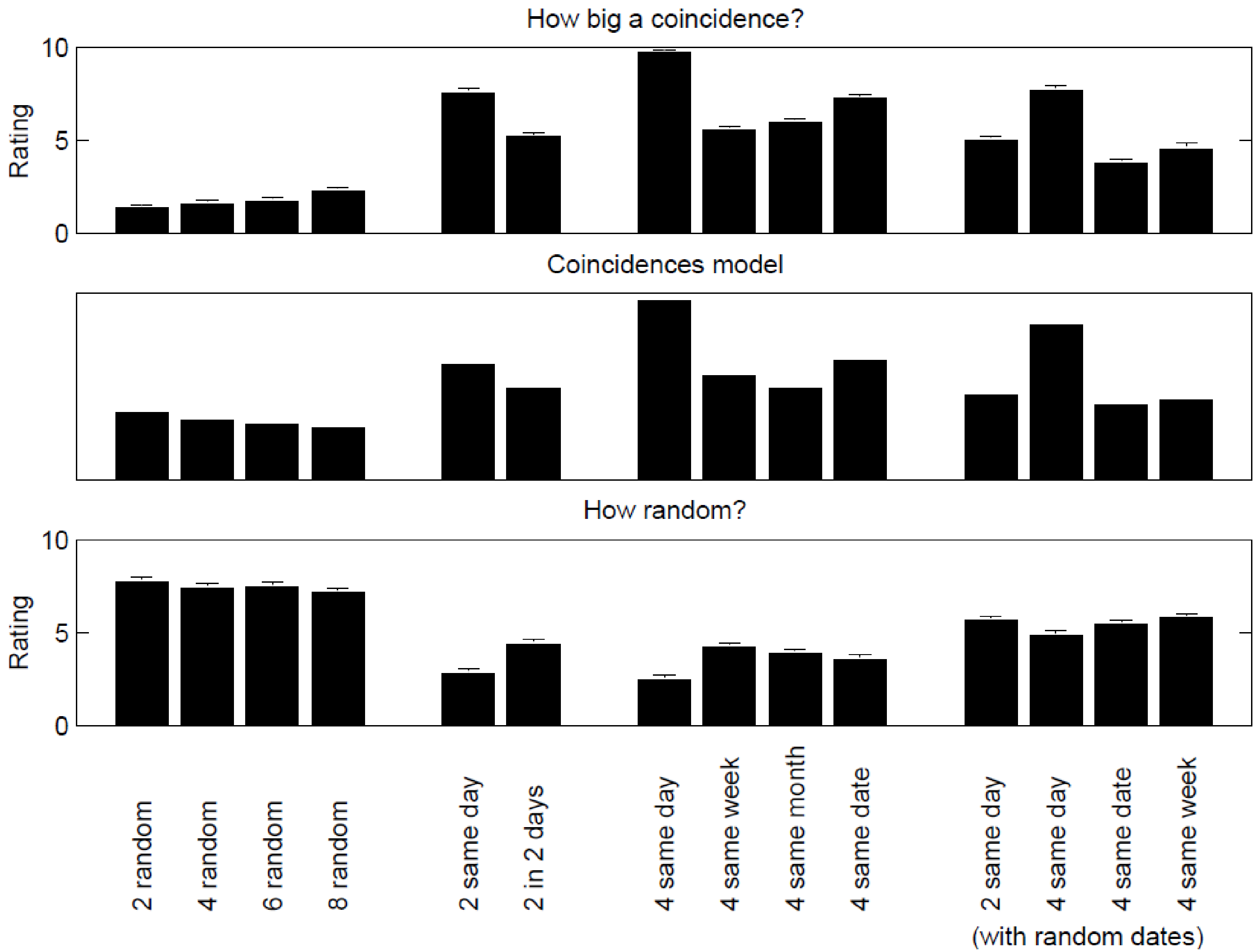
Coincidences, Randomness, and the Birthday Paradox

How big of a coincidence is it that four people have their birthdays on Oct 4, Oct 4, Oct 4, Oct 4 vs. May 14, Jul 18, Aug 21, and Oct 4?

Basically the same model, except now, regularities are related to patterns in months/dates.

Participants asked how big a coincidence and how random?

Coincidences, Randomness, and the Birthday Paradox



Conclusions

People's randomness judgments seem at odds with probability theory.

But there is systematicity to it…

If we reformulate the computational problem as randomness AS OPPOSED TO regular, we can capture human judgments.

It generalizes beyond just randomness, but also to coincidences.

Discussion Questions

- What is the correct normative problem?
- Does modeling randomness judgments even matter? (is this just some weird phenomenon?)
- Does this generalize to (any) real-world domains?
- What is the set of regularities? How do we learn them for a new domain?
- Are randomness judgments an inevitable consequence of an ideal observer *looking for* patterns in a noisy world? Or is it just irrational?